

3

OHM'S LAW

This chapter introduces resistance and Ohm's law – the most important mathematical law of all in the electrotechnology industry. It also explains how to work with very large and very small numbers using 'engineering notation' and the role of metric prefixes (multipliers) in writing large and small values.

Chapter outline

- 3.1 Resistance and conductance
- 3.2 Ohm's law
- 3.3 Metric prefixes
- 3.4 Scientific and engineering notation
- 3.5 Using engineering notation



3.1 Resistance and conductance

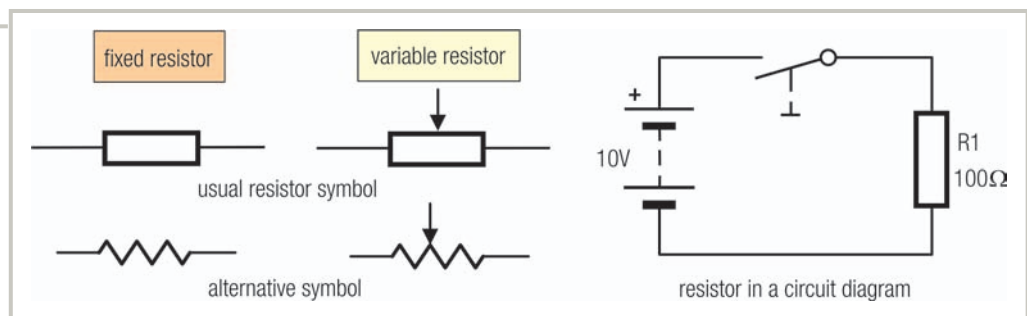
Chapter 1 explained that resistance in an electrical circuit is *opposition to the flow of current*. You also saw in Chapter 2 that resistance is present to some extent in all electrical conductors, depending on the material the conductor is made of. Resistance is, therefore, a fundamental property of an electric circuit. Chapter 5 looks in detail at resistance and in particular at a component called the *resistor*. This is an electrical component designed to have a certain value of resistance.

For the purposes of this chapter, you need to know the symbols used to depict resistance in an electric circuit, and that resistance is measured in *ohms*. Sometimes we use the Greek symbol Ω (*omega*) rather than write the word ohms. A resistance of 100 ohms is written as 100 Ω .

Although electrical cables have resistance, it's sometimes convenient to compare types of cables by their *conductance*, not their resistance. Electrical conductance is given the symbol G , and is a measure of the ability of a conductor to carry an electric current. Its value is the reciprocal of the cable's resistance and is measured in *siemens*, which has the symbol S . (Previously the unit for conductance was the *mho*, which is ohm spelt backwards.) A resistance of 10 ohms therefore equals a conductance of $1/10$ or 0.1 siemens. A resistance of 0.1 ohms equals 10 S . Notice that as resistance goes down in value, the conductance increases.

FIGURE 3.1

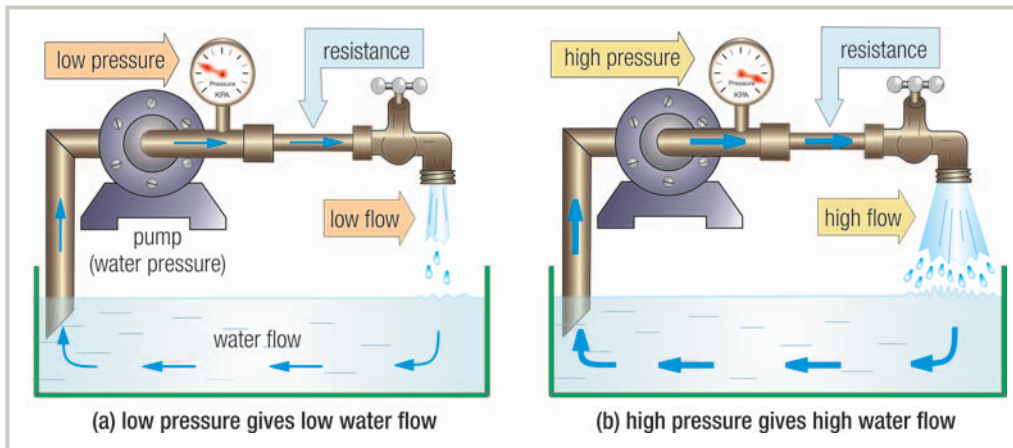
Resistor symbols, and how a resistor is depicted in a circuit diagram



3.2 Ohm's law

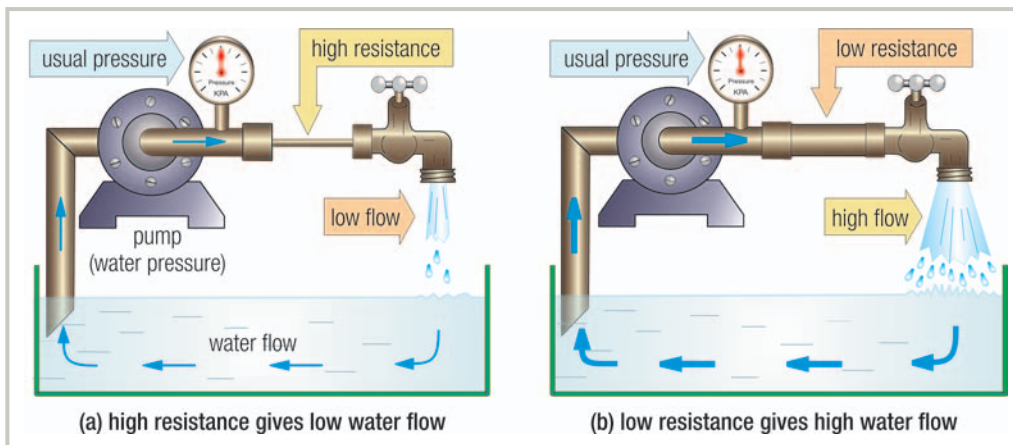
In 1827, Georg Simon Ohm (German physicist, 1787–1854) presented his laws on resistance, voltage and current. However, he received so much criticism that he spent the next six years as a little-known teacher in Berlin. Eventually his work was accepted and in 1881 a meeting of the International Electrical Congress in Paris named the unit of electrical resistance the 'ohm' in his honour.

Ohm's law brings together the three fundamental electrical quantities of voltage, current and resistance. The relationship between these three quantities can be shown with a water pump pushing water through a pipe. Figure 3.2 shows something you already know – if the pressure from a water pump is low, the water flow is also low. Increase the pressure, and the water flow increases.

**FIGURE 3.2**

Water flow is directly proportional to the water pressure

Figure 3.3 shows something just as obvious. If the pipe carrying the water has a restriction put in it (and the pump pressure stays the same), the water flow reduces. The restriction could be a tap, or as in the diagram, a piece of thin pipe.

**FIGURE 3.3**

Water flow is inversely proportional to resistance of the pipe

As described in Chapter 1, voltage is electrical pressure, represented by the water pressure in Figures 3.2 and 3.3. Current is a flow of electrons, represented here by the water flow. Therefore, because the water flow increases when the water pressure is increased and *decreases* when the restriction (or resistance) is increased, we can say that:

- 1 *Current is directly proportional to voltage.* That is, if the resistance doesn't change, the current will increase if the voltage increases, and decrease if the voltage is decreased.
- 2 *Current is inversely proportional to resistance.* That is, if the voltage stays the same, the current will decrease if the resistance of the circuit is increased, and increase if the resistance is decreased.

Putting this together, Ohm's law says:

Ohm's law

The current flowing between any two points in an electric circuit is directly proportional to the voltage between the two points, and is inversely proportional to the resistance of the circuit between the two points.

From this we can write the mathematical relationships between current (I) and voltage (V), and between current (I) and resistance (R):

$I \propto V$ (means I is *proportional to* V)

$I \propto \frac{1}{R}$ (means I is *inversely proportional to* R)

Ohm's law takes away the 'proportional to' ($\propto$) sign, giving us the most important equation in electrical theory:

Ohm's law for current

$$I = \frac{V}{R}$$

where:

I = current in amperes

V = voltage in volts

R = resistance in ohms

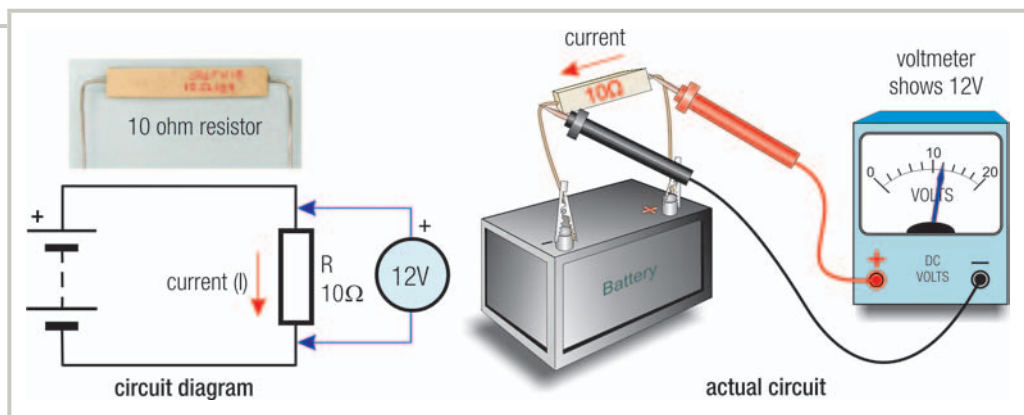
Using Ohm's law to find current

As you saw in Chapter 1, if you want to measure the current flowing in a circuit, it's usually necessary to break into the circuit so the current can also flow through the ammeter. This is often difficult, even impossible to do. An easier way is to calculate the value of the current with Ohm's law. You need to know, or be able to measure, the resistance and the voltage.

Figure 3.4 shows a 10 ohm resistor connected across a 12 V battery. In this diagram, the resistance value ($10\ \Omega$) is stamped on the resistor, and the voltage is measured with a voltmeter. Because the values of resistance and voltage are known, the current flowing through the resistor can be calculated.

FIGURE 3.4

The current in the circuit can be found with Ohm's law, because resistance and voltage are known



From the measured values in the circuit above, $R = 10\ \Omega$ and $V = 12\text{ V}$.

Therefore, by Ohm's law: $I = \frac{V}{R} = \frac{12}{10} = 1.2\text{ amperes}$

EXAMPLE 3.1

A voltage of 25 volts is measured across a resistor of 15 ohms. Calculate the current flowing in the resistor.

Solution

Values $V = 25\text{ volts}$
 $R = 15\text{ ohms}$
 $I = ?$





Equation $I = \frac{V}{R}$
 $I = \frac{25}{15}$

Answer **$I = 1.67$ amperes**

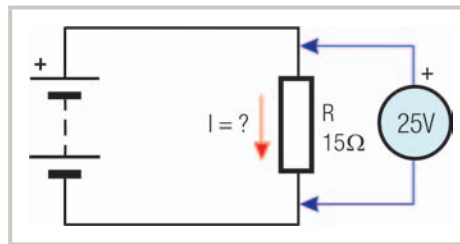


FIGURE 3.5

Circuit diagram for Example 3.1

Using Ohm's law to find voltage

In some cases you might know the values of the current and resistance in a circuit, but not the voltage. Because Ohm's law is a mathematical equation, it can be rearranged by transposing it to make voltage the *subject*. The subject is the unknown quantity and is on the left-hand side of the equal (=) sign, with the remaining terms on the right-hand side. The process for transposing the equation $I = \frac{V}{R}$ so V is the subject is shown in Figure 3.6.

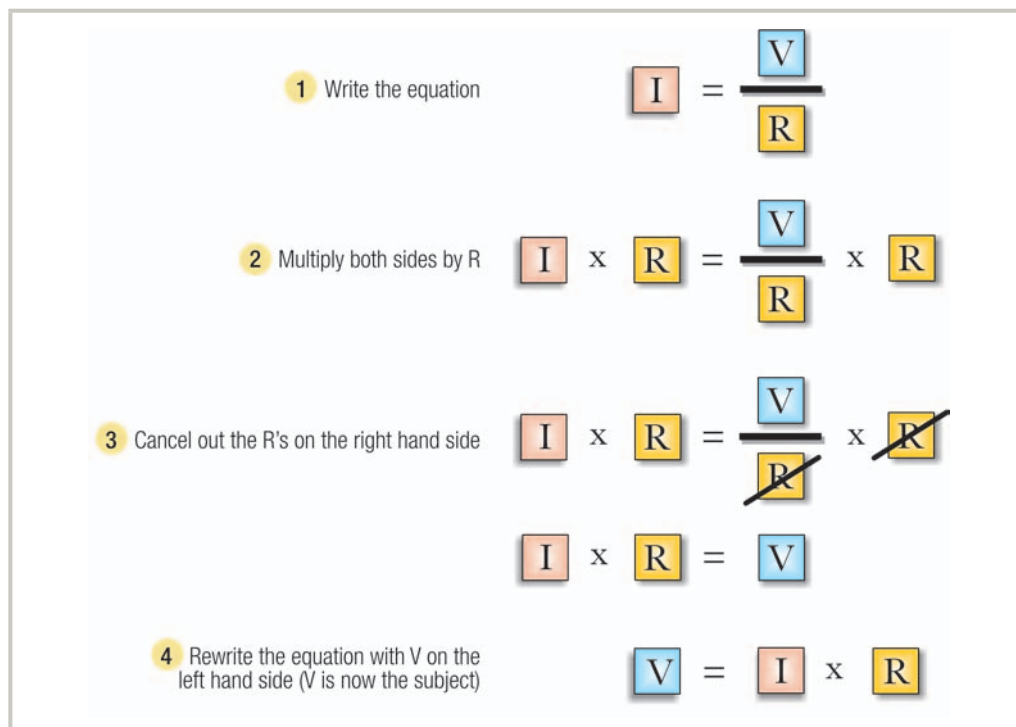


FIGURE 3.6

Transposing Ohm's law in terms of voltage (V)

The equation to find the voltage if current and resistance are known is:

Ohm's law for voltage

$$V = IR$$

where:

V = voltage in volts

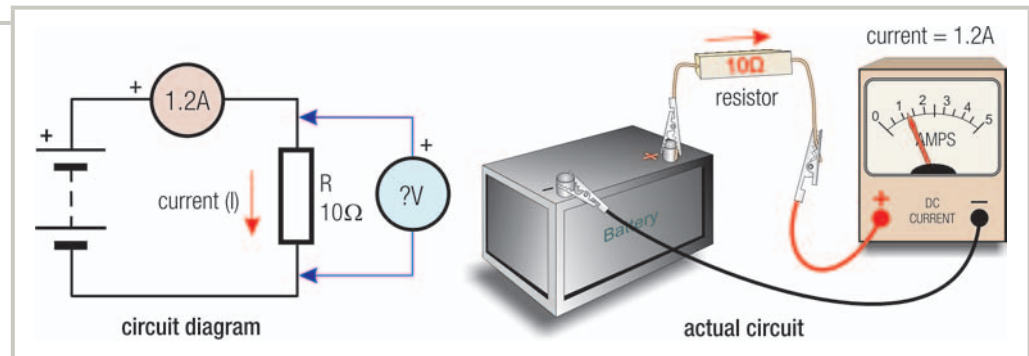
I = current in amperes

R = resistance in ohms

Figure 3.7 shows a 10 ohm resistor and an ammeter connected to a battery of unknown voltage. The ammeter is showing a current of 1.2 A, and the resistance value (10 Ω) is stamped on the resistor. Because the values of resistance and current are known, the voltage across the resistor can be calculated.

FIGURE 3.7

The battery voltage in the circuit can be found with Ohm's law, because resistance and current are known



From the measured values in the circuit in Figure 3.7, $R = 10 \Omega$ and $I = 1.2 \text{ A}$.

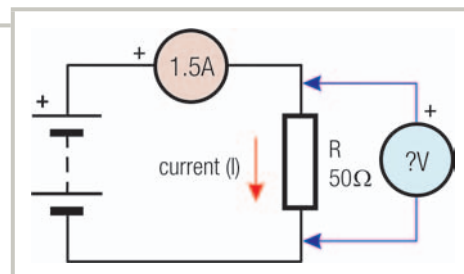
Therefore, by Ohm's law: $V = I \times R = 10 \times 1.2 = 12 \text{ volts}$. This is to be expected, as the circuit is the same as in Figure 3.4.

EXAMPLE 3.2

A 50 ohm resistor has a current of 1.5 amperes flowing through it. Calculate the voltage across the resistor.

FIGURE 3.8

Circuit diagram for Example 3.2

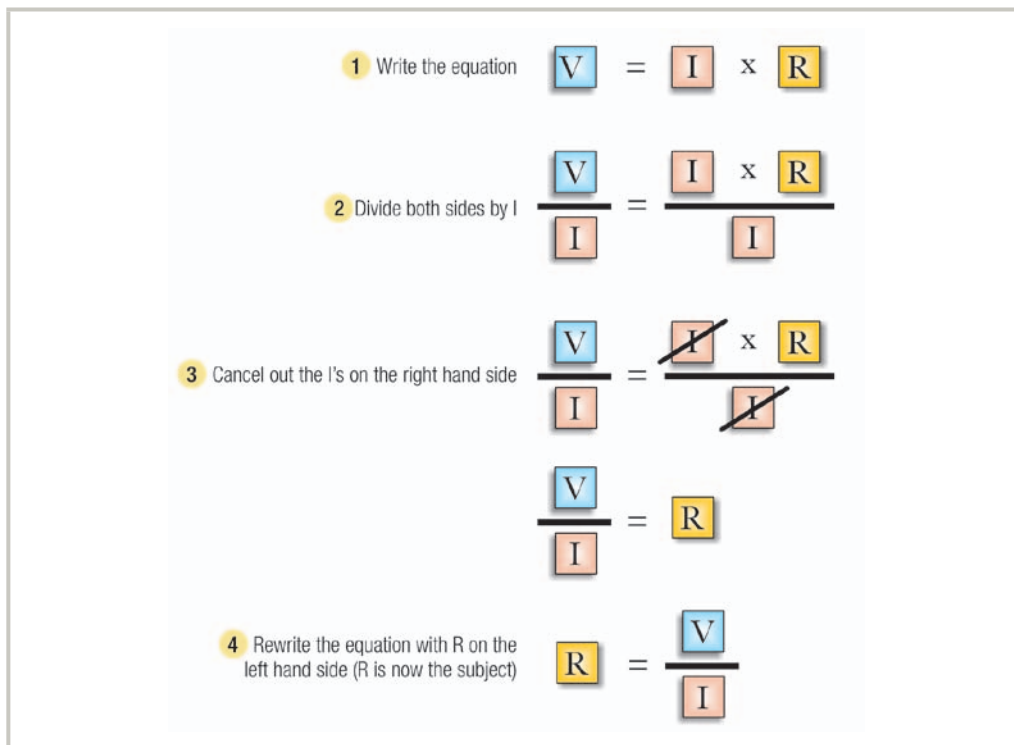


Solution

Values	$I = 1.5 \text{ amperes}$ $R = 50 \text{ ohms}$ $V = ?$
Equation	$V = IR$ $V = 1.5 \times 50$
Answer	$V = 75 \text{ volts}$

Using Ohm's law to find resistance

In some cases the voltage and current are known, with resistance the unknown quantity. The sequence to make R the subject of the equation is shown in Figure 3.9, in which you start with the equation $V = I \times R$.

**FIGURE 3.9**

The sequence to transpose $V = I \times R$ so R is the subject

The equation to find resistance if the voltage and the current are known is:

Ohm's law for resistance

$$R = \frac{V}{I}$$

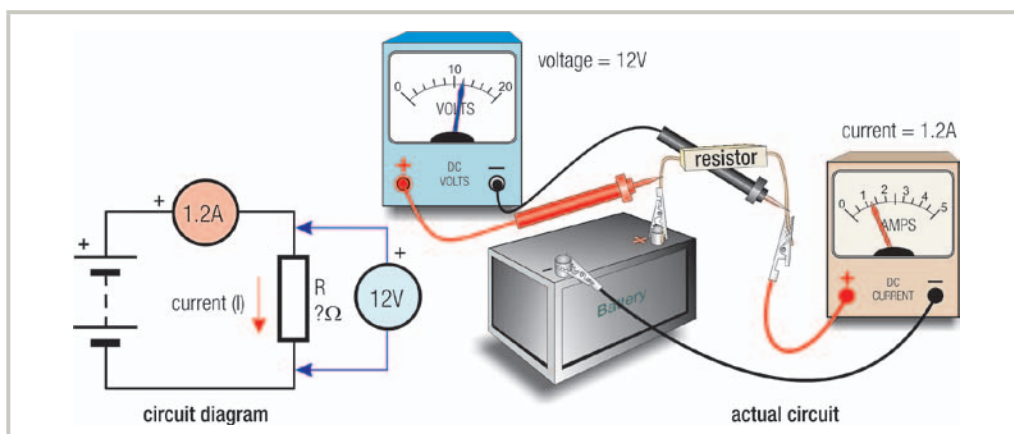
where:

V = voltage in volts

I = current in amperes

R = resistance in ohms

Figure 3.10 shows a resistor of unknown value connected to a battery that, according to the voltmeter, is producing 12 V across the resistor. The current in the resistor, as shown by the ammeter, is 1.2 A. Because the values of voltage and current are known, the resistance of the resistor can be calculated.

**FIGURE 3.10**

The unknown resistance value can be found with Ohm's law, because voltage and current are known

From the measured values in the circuit above, $V = 12\text{ V}$ and $I = 1.2\text{ A}$.

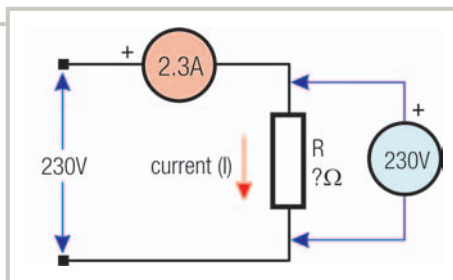
Therefore, by Ohm's law: $R = \frac{V}{I} = \frac{12}{1.2} = 10\text{ ohms}$. This is to be expected, as the circuit values are the same as in Figures 3.4 and 3.7.

EXAMPLE 3.3

An electric heater connected to 230 V takes a current of 2.3 amperes. Calculate the resistance of the heater.

FIGURE 3.11

Circuit diagram for Example 3.3



Solution

Values

$$V = 230\text{ volts}$$

$$I = 2.3\text{ amperes}$$

$$R = ?$$

Equation

$$R = \frac{V}{I}$$

$$R = \frac{230}{2.3}$$

Answer

$$R = 100\text{ ohms}$$

Sometimes E (for EMF) is given instead of V (for voltage). There is no difference between these two as far as the Ohm's law equations are concerned. If E is given, rather than V , substitute E for V in the equations. The equations then become:

$$I = \frac{E}{R} \text{ where } E \text{ is the applied EMF in volts}$$

$$E = IR$$

$$R = \frac{E}{I}$$

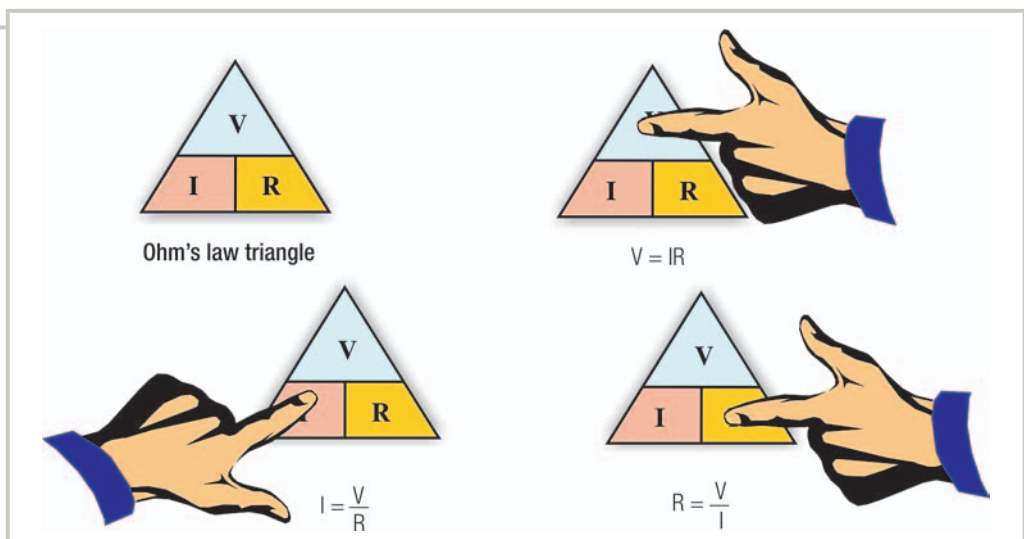
The three equations Ohm presented in 1827 are now seen as the most important equations of all in electrical and electronic theory. You need to know these equations and, of course, how to use them.

Ohm's law triangle

The triangle in Figure 3.12 might help you remember the three Ohm's law equations. To find any of the three equations, cover the value you want to find and the equation is the other two terms.

FIGURE 3.12

Ohm's law represented in a triangle. Cover the term you want to find: the equation is the other two terms.



EXERCISE 3.1

- 1 A resistor has 20 V across it when a current of 0.5 A is flowing through it. Find the value of the resistor.
- 2 What is the voltage across a 120 ohm resistor when a current of 0.25 A is flowing through it?
- 3 A length of cable has a resistance of 0.5 ohms. What is the voltage across the cable when a current of 20 A is flowing through the cable?
- 4 What value of resistance is needed to obtain a voltage across the resistor of 35 volts when the current in the resistor is 2.5 A?
- 5 A faulty switch has a resistance of 2.5 ohms. When the switch is turned on, a current of 8 A flows through the switch. Calculate the voltage across the switch.
- 6 A circuit has a resistance of 460 ohms. How much current flows in the circuit when it is connected to a voltage of 230 V?

3.3 Metric prefixes

In the examples given so far, the numbers have been simple to handle as they were neither very large nor very small. However, this is often not the case. For example, resistor values can range from 0.0001 ohm to 82 million ohms. The current flowing in a circuit can be from a few millionths of an ampere up to thousands of amperes, and power stations supply millions of watts of electrical power.

In everyday life, we refer to kilometres rather than saying one thousand metres, or millimetres instead of saying one thousandth of a metre. The terms 'kilo' and 'milli' are known as *metric prefixes* or *multipliers* and are given the symbols 'k' and 'm'. That is, rather than write 25 kilometres, you would write 25 km, or 66 mm instead of 66 millimetres. Table 3.1 lists the prefixes used in electrotechnology.

METRIC PREFIXES**TABLE 3.1**

Term	Symbol	Multiply by	Example
tera	T	1 000 000 000 000	2 THz = 2 000 000 000 000 hertz
giga	G	1 000 000 000	2 GW = 2 000 000 000 watt
mega	M	1 000 000	39 MΩ = 39 000 000 ohm
kilo	k	1000	33 kV = 33 000 volts
milli	m	0.001	2 mA = 0.002 amperes
micro	μ	0.000 001	4 μF = 0.000 004 farads
nano	n	0.000 000 001	9 nA = 0.000 000 009 amperes
pico	p	0.000 000 000 001	3 pA = 0.000 000 000 003 amperes

The four most commonly used prefixes in electrotechnology are mega, kilo, milli and micro. The two things you need to be able to do are listed below.

- 1 Remove a prefix from a number to get its base unit so the number can be used in an equation. For instance, you can't divide 2 V by 2 k ohms, instead you need to replace the k with its numerical value, which from Table 3.1 is 1000, giving a resistance of 2000 ohms. The base units for the three Ohm's law equations are volts, amperes or ohms.

- 2 Use a prefix to make a large or small number easier to understand. For example, rewriting 800 000 m as 800 km.

The following four examples show how to convert a value expressed with a prefix into a base unit.

EXAMPLE 3.4

How many volts are there in 3.9 MV?

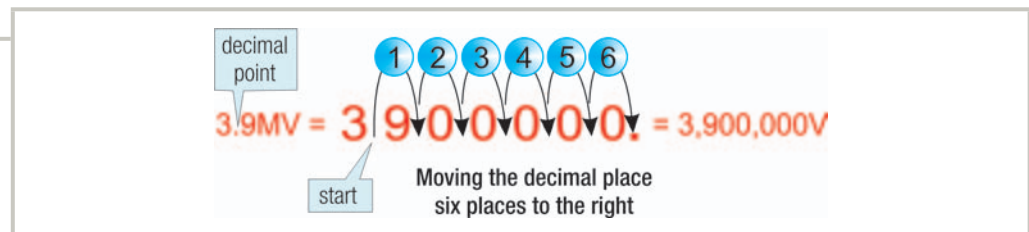
Solution

M (mega) means *multiply* the number 3.9 by 1 000 000, which is the same as moving the decimal place *six* places to the right. This process is shown in Figure 3.13.

Answer: **3 900 000 volts**

FIGURE 3.13

Converting 3.9 MV to volts by moving the decimal to the right



EXAMPLE 3.5

Convert 2.75 kV to volts.

Solution

k (kilo) means *multiply* the number by 1000. This is the same as moving the decimal place *three* places to the right.

Answer: **2750 volts**

Both of these conversions can be done on a calculator. For Example 3.4, enter 3.9 as the first value, then multiply it by 1 000 000. For Example 3.5, enter 2.75 then multiply it by 1000.

EXAMPLE 3.6

Convert 25.8 μ A to amperes.

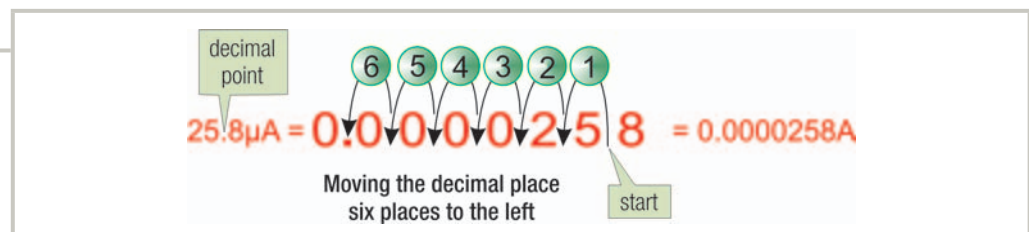
Solution

μ (micro) means *divide* the number by 1 000 000 or move the decimal place *six* places to the left, as shown in Figure 3.14.

Answer: **0.0000258 amperes**

FIGURE 3.14

Converting 25.8 μ A to amperes by moving the decimal point to the left



EXAMPLE 3.7

Write 4.76 mA as a value in amperes.

Solution

m (milli) means *divide* the number by 1000 which is the same as moving the decimal place *three* places to the left.

Answer: **0.00476 amperes**

Examples 3.6 and 3.7 can also be done on a calculator by dividing the value in Example 3.6 by 1 000 000 and dividing by 1000 in Example 3.7.

The next two examples show how to express a large or a small value with a prefix.

EXAMPLE 3.8

Convert 2560 000 ohms to $M\Omega$ (pronounced megohms).

Solution

Divide the value by 1 000 000 or move the decimal point *six* places to the left. Put the letter 'M' after the new value to show the missing decimal places.

Answer: **2.56 $M\Omega$**

EXAMPLE 3.9

Rewrite 0.0056 A to a value in milliamperes (mA).

Solution

Multiply the value by 1000 or move the decimal point *three* places to the right. Add the letter 'm' after the new value to show the missing decimal places.

Answer: **5.6 mA**

In these examples, you will have noticed that the prefix replaces a certain number of zeros. This works only if the zeros are either before or after the value. For instance, a value of 2000.007 can't be simplified with a prefix, as the zeros are between two numbers.

Table 3.2 summarises how to remove or add a prefix. It shows the number of places to move the decimal point, and which way to move it. Or if you are using a calculator, the table also shows whether you divide or multiply the value, and by how much. Only the four most commonly used prefixes are listed, but the same rules apply for all prefixes. You just need to know the number of decimal places or the number they represent.

TABLE 3.2

USING PREFIXES			
Prefix	Symbol	To add the prefix, move the decimal point:	To remove the prefix, move the decimal point:
mega	M	6 places to the left (or divide by 1 000 000) e.g. 1 000 000 Ω = 1 MΩ	6 places to the right (or multiply by 1 000 000) e.g. 1 MΩ = 1 000 000 Ω
kilo	k	3 places to the left (or divide by 1000) e.g. 1000 V = 1 kV	3 places to the right (or multiply by 1000) e.g. 1 kV = 1000 V
milli	m	3 places to the right (or multiply by 1000) e.g. 0.001 A = 1 mA	3 places to the left (or divide by 1000) e.g. 1 mA = 0.001 A
micro	μ	6 places to the right (or multiply by 1 000 000) e.g. 0.000 001 A = 1 μA	6 places to the left (or divide by 1 000 000) e.g. 1 μA = 0.000 001 A

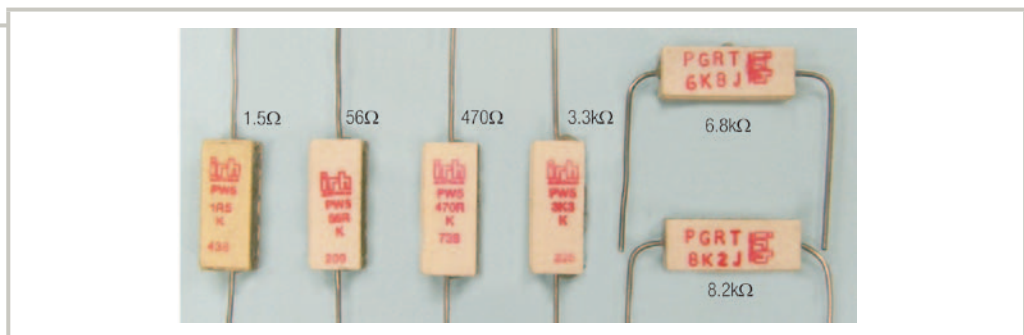
Decimal point notation

Values expressed with a prefix often have a decimal point. For example, 6 800 000 ohms converted to megohms gives 6.8 M Ω . However, on a circuit diagram or the component itself, the decimal point could be badly printed, even missing. To avoid the problem, the prefix symbol is sometimes placed where the decimal point would otherwise be. So another way of writing 6.8 M Ω is 6M8 Ω . A resistance value of 2700 Ω can be written as 2k7 Ω or 2.7 k Ω . A current of 25.6 μ A could be written as 25 μ 6A and a voltage of 30.5 mV as 30m5V.

Figure 3.15 shows a range of resistors where the decimal point is replaced by a prefix. Where a prefix is not required, the decimal point is replaced with the letter R (for resistance). There is no standard method. The important thing is to be careful when writing a value with a decimal point, so that anyone reading it has no doubt. For example, write 0.005 instead of .005, as the leading zero helps show there's a decimal point.

FIGURE 3.15

Resistance values where a letter replaces the decimal point



EXERCISE 3.2

- 1 Use a prefix to write 1 200 000 ohms in a simpler way.
- 2 Simplify 0.000056 A by expressing the number with a prefix.
- 3 How many volts does 3.45 kV represent?
- 4 Convert 6530 millivolts to volts.

- 5 A thermocouple produces 30 mV. Express this value in volts.
- 6 A resistance value is given as 0.000003 ohms. Express this value in micro ohms.
- 7 Express 0.04 V in millivolts (mV).
- 8 Electric power is transmitted at 330 kV. How many volts does this represent?
- 9 Express 0.00568 watts in milliwatts (mW) and also in microwatts (μ W).
- 10 Convert the resistance value of 27k6 ohms to ohms.

3.4 Scientific and engineering notation

Metric prefixes, or multipliers, are a convenient way to express large and small numbers but, as already explained, they need to be removed when used in a mathematical equation. However, another way is to express the prefix in mathematical form using *scientific notation*, in which each prefix is treated as a power of 10.

The term *power of 10* refers to the number of times 10 is multiplied by itself. For example, 1000 equals $10 \times 10 \times 10$. That is, 10 is multiplied by itself three times, or raised to the power of 3. So, rather than write $10 \times 10 \times 10$, we use the notation 10^3 . The power (3 in this case) is called the *exponent*.

Scientific notation also gives an easy way of comparing magnitudes, in which a change by 'an order of magnitude' means a change by a factor of 10. For example, if the current in a circuit is 40 A, and it increases by an order of magnitude, the current will be 400 A. A change from 40 A to 4000 A is two orders of magnitude. In scientific notation, 40 A is written as 4×10^1 and 4000 A becomes 4×10^3 . The change in magnitude is shown by the difference between the exponents, which in this case is $3 - 1 = 2$.

You need a scientific calculator to handle exponents and scientific notation. Figure 3.16 shows a basic scientific calculator, and how to enter the value 2×10^3 . When the = key is pressed, the display should show 2000.

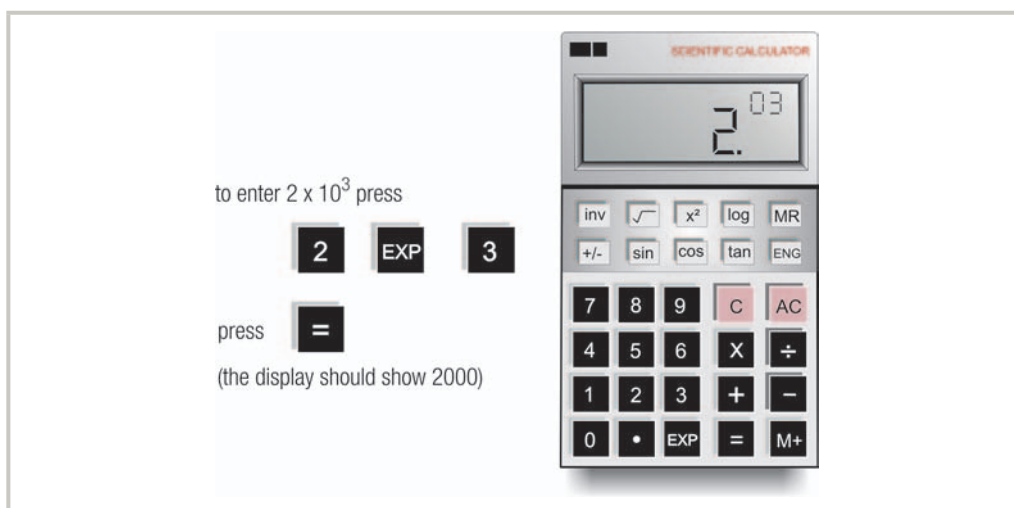


FIGURE 3.16

A scientific calculator can handle powers of 10. Press the EXP key (could be called EE or 10X, and could be anywhere on the keyboard) to enter a power of 10.

Each prefix used in the metric system represents a power of 10 and can be written in scientific notation. Therefore, because kilo (k) is 10^3 , to enter a value like 2 k Ω into a scientific calculator, follow the procedure shown in Figure 3.16, where after entering the value 2, the 'k' is entered as exponent 3. Table 3.3 shows the scientific notation for the prefixes used in electrical work.

TABLE 3.3

SCIENTIFIC NOTATION				
Term	Symbol	Multiply by	Scientific notation	Example
tera	T	1 000 000 000 000	10^{12}	$1.6 \text{ THz} = 1.6 \times 10^{12} \text{ Hz}$
giga	G	1 000 000 000	10^9	$4 \text{ GW} = 4 \times 10^9 \text{ W}$
mega	M	1 000 000	10^6	$3 \text{ M}\Omega = 3 \times 10^6 \Omega$
kilo	k	1000	10^3	$2.2 \text{ kV} = 2.2 \times 10^3 \text{ V}$
milli	m	0.001	10^{-3}	$5.6 \text{ mA} = 5.6 \times 10^{-3} \text{ A}$
micro	μ	0.000 001	10^{-6}	$89 \mu\text{A} = 89 \times 10^{-6} \text{ A}$
nano	n	0.000 000 001	10^{-9}	$6 \text{ nA} = 6 \times 10^{-9} \text{ A}$
pico	p	0.000 000 000 001	10^{-12}	$5 \text{ pA} = 5 \times 10^{-12} \text{ A}$

Engineering notation

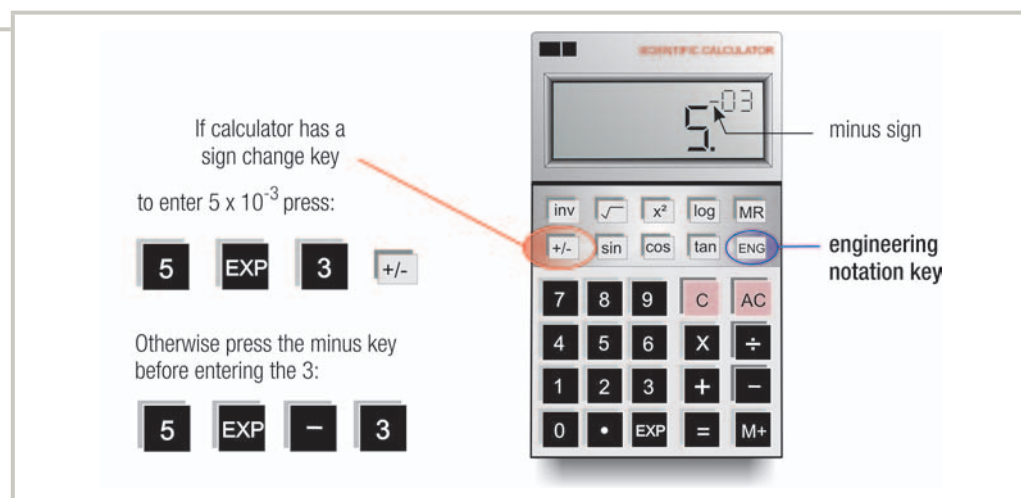
Notice that the exponents in Table 3.3 are all multiples of 3. While scientific notation can have any value for the exponent, engineering notation limits the exponent to a value that is a multiple of 3, such as 3, 6, 9 or 12. Therefore, terms such as 'centi' (10^2) and 'deci' (10^{-1}) are not used. Some calculators have a key marked ENG, which when pressed converts a value to engineering notation. For example, if the result of a calculation is 2 340 000, pressing the ENG key will display 2.34×10^6 . Pressing the ENG key again will show 2340×10^3 and so on. Engineering notation is used in electrical calculations.

Values less than 1 have a negative exponent. You can see in the table that milli (one thousandth) equals 10^{-3} and micro equals 10^{-6} . In fact, 10^{-3} equals 1 divided by 10^3 which, as the table shows, gives 0.001. If you enter 1 divided by 1000 into a scientific calculator, the answer might be shown as 1^{-03} , which is how many calculators display 10^{-3} . Some types of calculators will show the result as 0.001. Similarly, 10^{-6} equals 0.000 001, which is the same as 1 divided by 10^6 .

Figure 3.17 shows how to enter a negative exponent into a calculator and what the display should show when 5×10^{-3} is entered. Incidentally, 5×10^{-3} equals 0.005.

FIGURE 3.17

Entering a negative exponent into a calculator



3.5 Using engineering notation

Prefixes make large or small numbers easier to write, and scientific notation makes large or small values easier to handle mathematically. As you've seen, if a value is written with a prefix (e.g. 33 kV) then to use the value in an equation, the prefix is replaced by its equivalent scientific notation (e.g. 10^3), as listed in Table 3.3.

In electrical work, you will need to be able to multiply and divide numbers expressed in scientific notation, and also to add and subtract them. When a value is expressed with scientific notation, the value has two components: the *coefficient* and the *exponent*, as shown in Figure 3.18. These two terms are now used to describe how to work with numbers expressed in scientific notation.

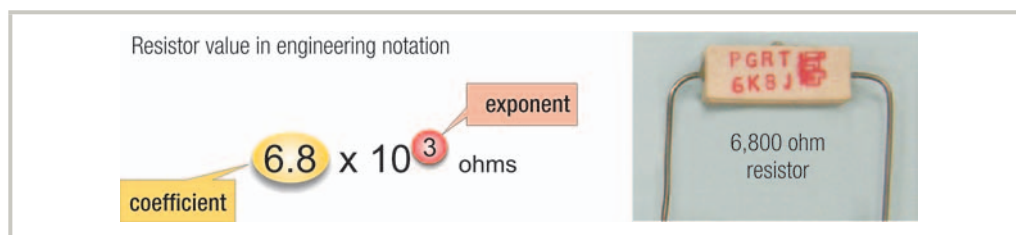


FIGURE 3.18

Exponents and coefficients

Multiplication

For multiplication – *multiply the coefficients and add the exponents*. Adding the exponents must be done algebraically where adding a minus number means you subtract that number.

For example:

$$(3 \times 10^3) \times (4 \times 10^6) = 12 \times 10^9 \text{ (because 3 plus 6 gives 9)}$$

$$(15 \times 10^6) \times (5 \times 10^{-3}) = 75 \times 10^3 \text{ (because +6 added to -3 gives +3)}$$

$$(2.2 \times 10^{-6}) \times (1.5 \times 10^{-3}) = 3.3 \times 10^{-9} \text{ (because -6 added to -3 gives -9)}$$

EXAMPLE 3.10

Find the voltage across a $1.5 \text{ M}\Omega$ resistor that has a current of 5 mA flowing through it. Express the voltage value with a prefix.

Solution

Values $V = ? \text{ volts}$
 $I = 5 \text{ mA}$
 $R = 1.5 \text{ M}\Omega$

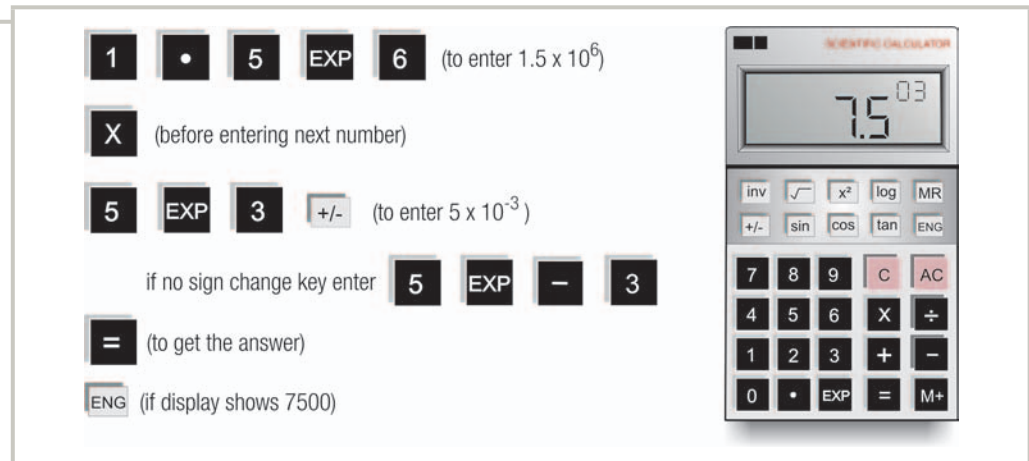
Equation $V = IR$
 $V = (1.5 \times 10^6) \times (5 \times 10^{-3})$ [replacing prefixes with engineering notation]
 $= 1.5 \times 5 \times 10^3$ [adding the exponents in which +6 added to -3 gives +3]
 $= 7.5 \times 10^3 \text{ volts}$

Answer **$V = 7.5 \text{ kilovolts or } 7.5 \text{ kV}$**

The sequence to use for a calculator is shown in Figure 3.19. If the answer is displayed as 7500, press the ENG key to change the display to that shown in the diagram. Repeated pressing of the ENG key will change the exponent.

FIGURE 3.19

Calculator sequence to solve Example 3.10



Division

For division – divide the coefficients, then algebraically subtract the exponent of the number below the line (the divisor) from the exponent of the number above the line (the dividend). For example:

$$\frac{6.6 \times 10^6}{2 \times 10^3} = 3.3 \times 10^3$$

$$\frac{8 \times 10^3}{4 \times 10^6} = 2 \times 10^{-3} \text{ (6 subtracted from 3 gives } -3\text{)}$$

$$\frac{15 \times 10^6}{3 \times 10^{-3}} = 0.5 \times 10^9 \text{ (subtracting a minus gives a plus, giving } 6 + 3 = 9\text{)}$$

Again, subtracting the exponents can be done mentally and the coefficients divided on a calculator.

EXAMPLE 3.11

When a voltage of 1.5 MV is applied to a circuit, a current of 3 mA flows. Find the resistance of the circuit and express its value with a prefix.

Solution

Values $V = 1.5 \text{ MV}$
 $I = 3 \text{ mA}$
 $R = ?$

Equation $R = \frac{V}{I}$

$$R = \frac{1.5 \times 10^6}{3 \times 10^{-3}} \text{ [replacing prefixes with engineering notation]}$$

$$R = \frac{1.5 \times 10^9}{3} \text{ [simplifying exponents]}$$

$$R = 0.5 \times 10^9 \text{ ohms}$$

$$R = 500 \times 10^6 \text{ ohms}$$

Answer **R = 500 megohms**

The calculator sequence for this example is shown in Figure 3.20. If your calculator gives the display shown, try pressing the ENG key again, which will usually change the exponent to the next engineering notation value (in this case to -6 , with a value of 500 for the coefficient, as in the answer to the example).

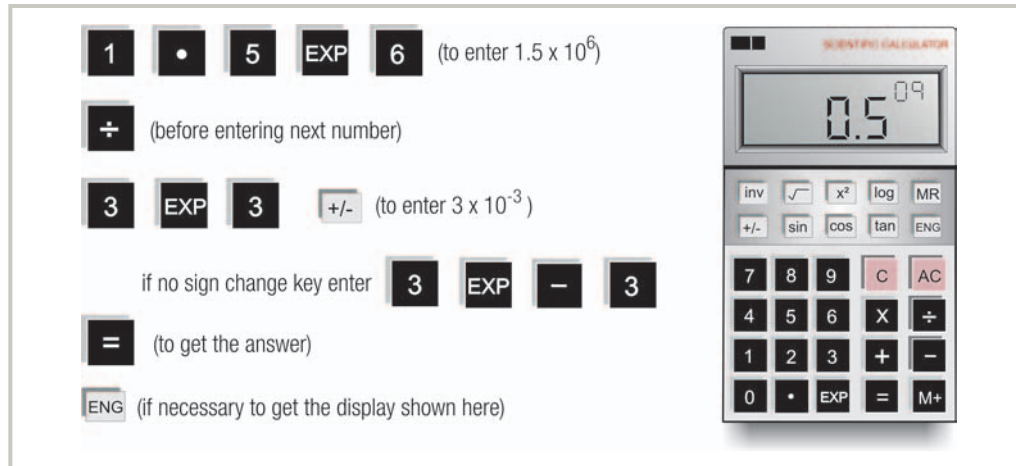


FIGURE 3.20

Calculator sequence to solve Example 3.11

Addition and subtraction

To add (or subtract) values written in engineering notation, the values must have the *same* exponent. The coefficients are added (or subtracted) and the exponent is left alone. For example, where the exponents are the same:

$$(3.5 \times 10^3) + (4.1 \times 10^3) = 7.6 \times 10^3$$

$$(45.6 \times 10^{-3}) + (6.2 \times 10^{-3}) = 51.8 \times 10^{-3}$$

$$(21.6 \times 10^{-6}) - (5 \times 10^{-6}) = 16.6 \times 10^{-6}$$

$$(11.4 \times 10^3) - (7.2 \times 10^3) = 4.2 \times 10^3$$

Where the values have different exponents, one of them has to have its decimal point shifted so the exponents are the same throughout. The four basic rules to change from one exponent to another are:

- 1 To *reduce a positive exponent, move the decimal point to the right*. For example, to convert 12 MV to kilovolts, write 12 MV in engineering notation (12×10^6 V), then move the decimal point three places right to give $12\,000 \times 10^3$ V (or 12 kV).
- 2 To *increase a positive exponent, move the decimal point to the left*. For example, to convert 1234 k ohms to megohms, write the value in engineering notation (1234×10^3 ohms then move the decimal point three places left to give 1.234×10^6 ohms or 1.234 MΩ).
- 3 To *increase a negative exponent, move the decimal point to the right*. For example, to convert 0.12 mA to microamps, write the value in engineering notation (0.12×10^{-3} A), then move the decimal point three places right to give 120×10^{-6} A (or 120 μA).
- 4 To *reduce a negative exponent, move the decimal point to the left*. For example, to convert 1234 μA to milliamps, write the value in engineering notation (1234×10^{-6} A), then move the decimal point three places left to give 1.234×10^{-3} A (or 1.234 mA).

These rules cover all possibilities and show not only how to change an exponent, but how to convert from one prefix to another.

EXAMPLE 3.12

Add 876 μA and 4.6 mA.

Solution

One of the values must be converted to have the same prefix (or exponent) as the other value. In this case we'll convert the microamp value to milliamps. This means we need to reduce the exponent from -6 to -3 by moving the decimal point *left* by three places:

Convert 876 μA to milliamperes (mA):

$$\begin{aligned} 876 \mu\text{A} &= 876 \times 10^{-6} \text{ A (as } 10^{-6} = \mu) \\ &= 87.6 \times 10^{-5} \text{ A (1 place left)} \\ &= 8.76 \times 10^{-4} \text{ A (2 places left)} \\ &= 0.876 \times 10^{-3} \text{ A (3 places left)} \end{aligned}$$

$$\mathbf{876 \mu\text{A} = 0.876 \text{ mA (as } 10^{-3} = \text{m})}$$

Do the addition:

$$4.6 \text{ mA} + 0.876 \text{ mA [in engineering notation} = (4.6 \times 10^{-3}) + (0.876 \times 10^{-3})]$$

$$\text{Answer} = 5.476 \text{ mA}$$

EXAMPLE 3.13

Subtract 120 k Ω from 3.9 M Ω .

Solution

One of the values must be converted to have the same prefix (or exponent) as the other value. In this case we'll convert 3.9 M Ω to its equivalent value in k Ω . This means we need to reduce the exponent from 6 to 3 by moving the decimal point *right* by three places.

Convert 3.9 megohms (M Ω) to kilo ohms (k Ω):

$$\begin{aligned} 3.9 \text{ M}\Omega &= 3.9 \times 10^6 \Omega \text{ (as } 10^6 = \text{M)} \\ &= 39 \times 10^5 \Omega \text{ (1 place right)} \\ &= 390 \times 10^4 \Omega \text{ (2 places right)} \\ &= 3900 \times 10^3 \Omega \text{ (3 places right)} \end{aligned}$$

$$\mathbf{3.9 \text{ M}\Omega = 3900 \text{ k}\Omega \text{ (as } 10^3 = \text{k})}$$

Do the subtraction:

$$3900 \text{ K}\Omega - 120 \text{ k}\Omega \text{ [in engineering notation} = (3900 \times 10^3) - (120 \times 10^3)]$$

$$\text{Answer} = \mathbf{3780 \text{ k}}$$

EXERCISE 3.3

- 1 Convert 200 μA to milliamps.
- 2 Express 0.4 GW in megawatts.
- 3 A current is displayed on an ammeter as 0.24 mA. What is this value in micro amperes (μA)?
- 4 Convert 660 kV to megavolts (MV).
- 5 Convert 0.059 mA to microamps (μA).
- 6 Express the resistance value of 330 k ohms in megohms.
- 7 Find the voltage across a 2.2 k Ω resistor that has a current of 3 mA flowing through it.
- 8 Calculate the current flowing in an 8.2 k Ω resistor if the voltage across the resistor is 10V.
- 9 Add these three resistor values together: 330 Ω , 33 k Ω , 3.3 M Ω .
- 10 Subtract 250 MW from 4.5 GW.

CHAPTER SUMMARY

- Resistance is opposition to current flow.
- Resistance has the symbol R , and is measured in ohms, which has the Greek symbol Ω .
- Conductance is the reciprocal (inverse) of resistance, and is a measure of how well a conductor can carry an electric current.
- Conductance has the symbol G , and is measured in siemens, symbol S .
- Ohm's law mathematically relates the three fundamental electrical quantities of voltage (V), current (I) and resistance (R).
- Ohm's law says that the current flowing between any two points in an electric circuit is directly proportional to the voltage between the two points, and is inversely proportional to the resistance of the circuit between the two points.
- The three Ohm's law equations are: $I = \frac{V}{R}$, $V = IR$ and $R = \frac{V}{I}$, where V = voltage in volts, I = current in amperes and R = resistance in ohms.
- Metric prefixes are used to abbreviate the written form of a large or small number. The most commonly used prefixes in electrotechnology are kilo (k), mega (M), milli (m) and micro (μ). Others include giga, pico and nano.
- A metric prefix symbol is sometimes used to replace the decimal point in a value.
- Scientific notation expresses a metric prefix in mathematical form, in which each prefix is treated as a power of 10.
- Engineering notation is a subset of scientific notation, and uses only powers of 10 that are multiples of 3.
- Problems involving values expressed in scientific notation can be solved with a scientific calculator, in which each entered value has two components: the *coefficient* (a number) and the *exponent* (power of 10).
- When multiplying: multiply the coefficients and algebraically add the exponents. Adding the exponents can be done mentally, the coefficients multiplied on a calculator.
- For division: divide the coefficients, then algebraically subtract the exponent of the divisor (the number below the line) from the exponent of the dividend.
- To add (or subtract) values written in scientific notation, the values must have the *same* exponent.

SELF-CHECK

We recommend doing the exercises first. Then answer these questions and check your answers at the back of the book.

- Define the term 'resistance'.
- Explain the term 'conductance'.
- What is meant by 'current is proportional to voltage'?
- If the resistance of a circuit is doubled and the voltage remains the same, what happens to the current in the circuit?
- A conductor has a resistance of 2 milliohms. What is its conductance?
- During operation, an electric toaster takes a current of 5 A from a 230 V power outlet. Calculate the resistance of the toaster.
- The resistance of a relay coil is 150 Ω . Determine the value of the current taken by the coil when it's connected to 12 V. Express the value with a metric prefix.
- A faulty electrical connection in a circuit has developed a resistance of 0.6 Ω . If the current through the connection is 50 A, calculate the voltage that will be across the connection.
- Solve the following:
 - $2 \times 10^{-3} =$
 - $(2.2 \times 10^3) \times (4 \times 10^{-3}) =$
 - $2.3 \times 10^{-6} =$
- Write the following electrical values in engineering notation:
 - 2 mA
 - 6.67 μ V
 - 4.7 M Ω
- Convert:
 - 820 k Ω to M Ω
 - 0.087 mA to μ A
 - 692 μ V to mV
- Find the total current of 250 μ A + 4.5 mA + 0.05 A. Express the answer in milliamps.

- 13** Convert:
- a** 0.0012 volts to millivolts
 - b** 0.000 009 6 amperes to μA
 - c** 35.6 mA to amperes
- 14** How much voltage is developed across a resistor of $56\text{ k}\Omega$ that has a current of $120\mu\text{A}$ flowing through it?
- 15** A circuit with a resistance of $450\ \Omega$ is connected to a voltage supply of 330 kV. Calculate the current flowing in the circuit.
- 16** The voltage across a resistor increases by an order of magnitude. If the current in the resistor is now 650 milliamps, what was it before the voltage increased?
- 17** How much current flows in a resistor of $1\text{ k}\Omega$ that has 10 V across it? Express the answer in milliamps.
- 18** The resistance of a heater element, when cold, equals $20\ \Omega$. When the element warms up, its resistance is $50\ \Omega$. If the heater is connected to 230 V, calculate the current the heater element takes when cold, and again when it's warmed up.
- 19** A conductor carrying a current of 120 A has a resistance of $0.07\ \Omega$. Calculate the voltage developed across the length of the conductor.
- 20** In an electrolysis tank, when 2.5 volts is applied between the anode and cathode, a current of 10 kA passes through between the electrodes via the electrolyte. Calculate the electrical resistance of the electrolyte. Express the resistance value in engineering notation.

ONLINE RESOURCES

Try Googling the following search terms:

- scientific notation (topic is covered well by many educational (.edu) websites)
- metric prefix (covered by numerous websites, Wikipedia lists all the currently used prefixes and their value in scientific notation)
- Ohm's law (numerous websites present Ohm's law, but most at too great a depth at this stage. Worth browsing).